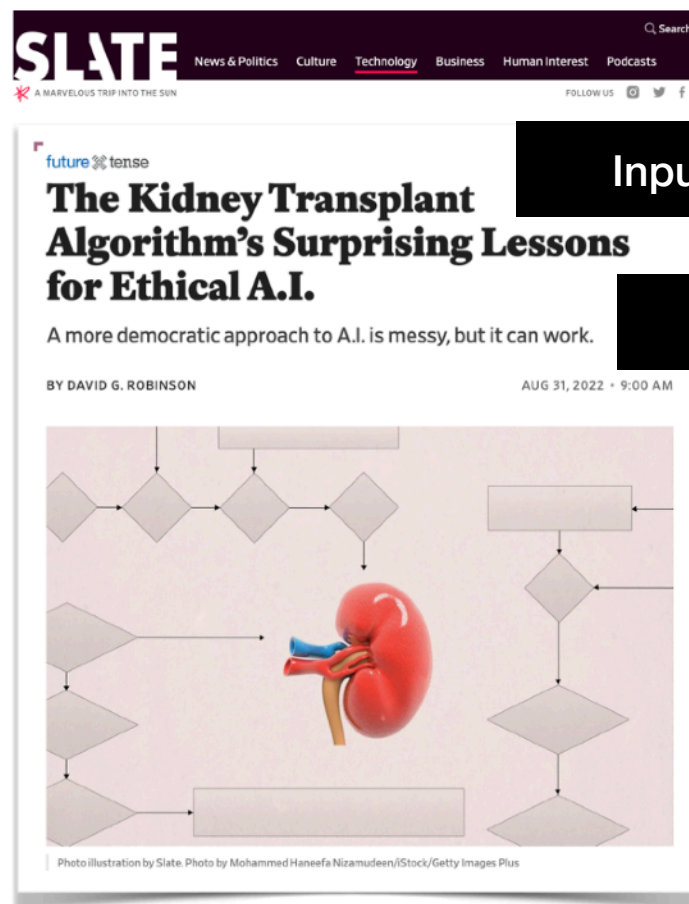




Deep Learning
What is this?

Introduction

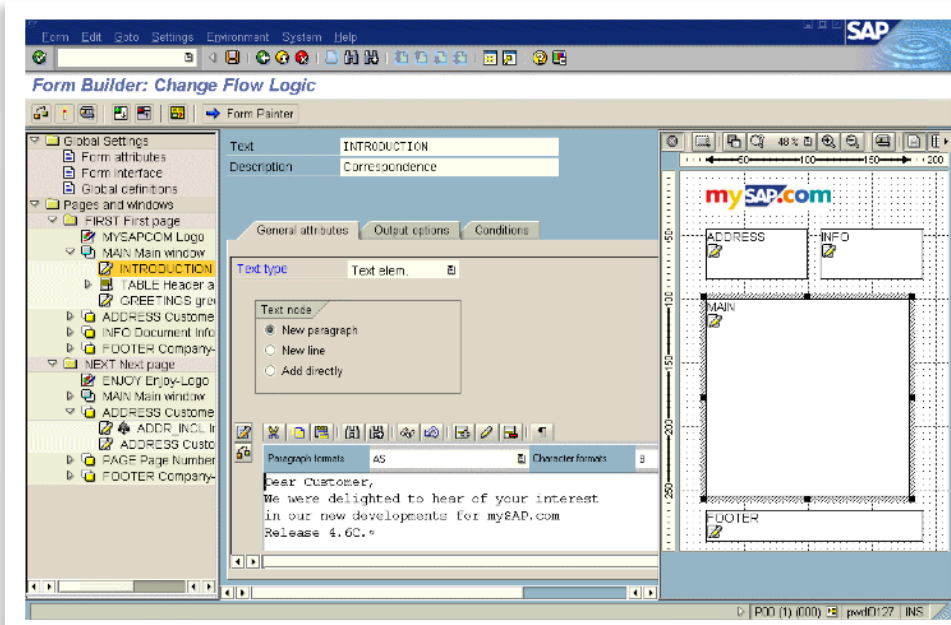
Until recently, most algorithms (including complex computer programs) you encountered in your daily life were created by humans.



Input data

Output data

```
main.py
1 def squareRoot(s):
2     x0 = s/2
3     difference = 1
4     accuracy = 0.01
5     while difference > accuracy :
6         x1 = 0.5*(x0+s/x0)
7         print(x1)
8         difference = x0-x1
9         print(difference)
10        x0 = x1
11    return round(x0,3)
12 print(squareRoot(26))
```



Introduction

All this problems could be solved by writing a series of rules that **determine the appropriate action for every possible situation the program might encounter.**

The ability to **manually** design automated systems that power functioning products and **handle various scenarios, often in new and unique situations**, is a remarkable cognitive achievement.

If you can consistently create solutions that work most of the time, you generally don't need to concern yourself with machine learning.

Introduction

But imagine you are tackling one of the following problems:

- Write a program that **predicts tomorrow's weather** given geographic information, satellite images, and a trailing window of past weather.
- Write a program that takes in a **factoid question**, expressed in free-form text, and answers it correctly.
- Write a program that, given an image, **locates every person** depicted in it and draws outlines around each.
- Write a program that presents users with **products that they are likely to enjoy** but unlikely, in the natural course of browsing, to encounter.

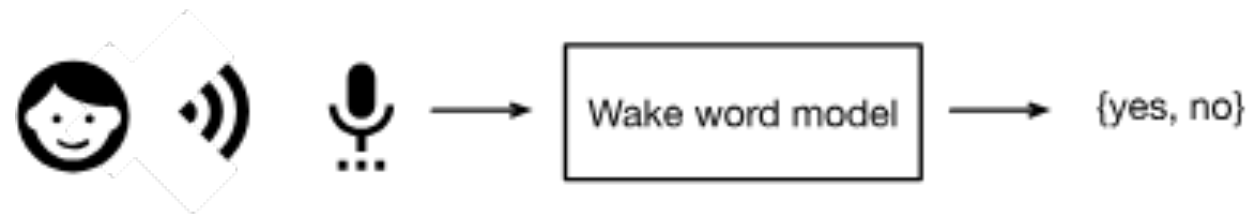
Introduction

Machine learning is the study of algorithms that can learn from experience (data).

As a machine learning algorithm accumulates more experience, typically in the form of **observational data** or **interactions with an environment**, its performance can improve.

Contrast this with a deterministic e-commerce platform, which follows the same business logic, no matter how much experience accrues, until the developers themselves learn and decide that it is time to update the software.

Models and parameters



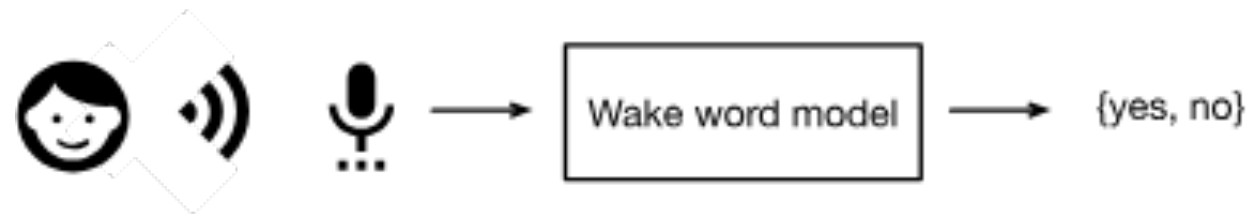
Imagine just writing a program to respond to a wake word such as “*Alexa*”, “*OK Google*”, and “*Hey Siri*”.

We can collect a huge **dataset** containing examples of audio snippets and associated labels, indicating which snippets contain the wake word.

In the currently dominant approach to machine learning, rather than explicitly designing a system to recognize wake words, we define a **flexible program** whose behavior is governed by a set of **parameters**. This program is capable of recognizing **patterns within the data**.

We then use the dataset to determine the optimal parameter values—those that enhance the program's performance for the specific task at hand.

Models and parameters



You can think of the **parameters** as knobs that we can turn, manipulating the behavior of the program.

Once the parameters are fixed, we call the program a **model**.

The set of all distinct programs (input–output mappings) that we can produce just by manipulating the parameters is called a **family** of models.

And the “meta-program” that uses our dataset to choose the parameters is called a **machine learning algorithm**.

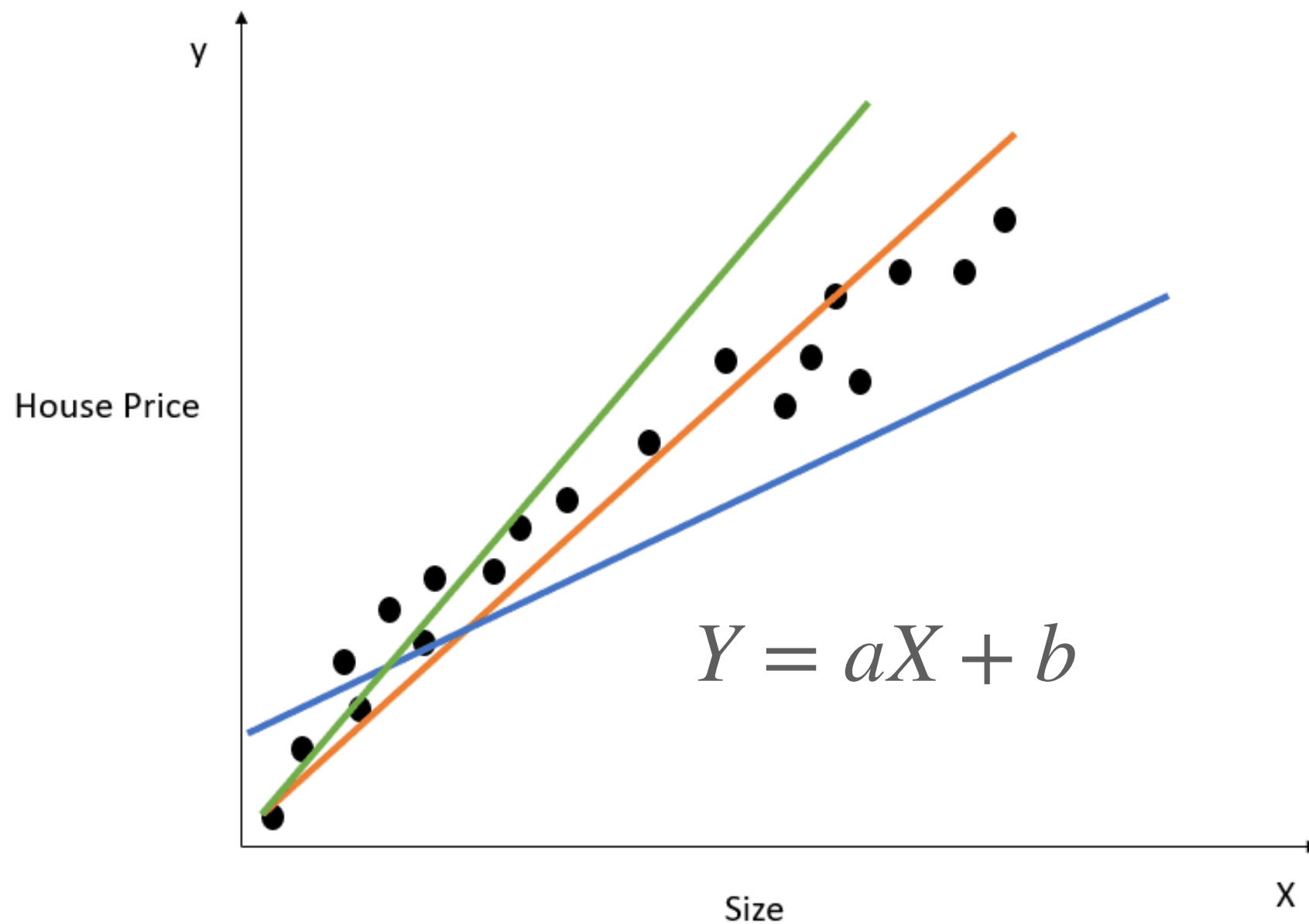
Models and parameters

Family: Linear Models, $y = ax + b$

Parameters: a and b

Learning Algorithm: Least Squares Minimization

Model: $y = 3.5x - 2$



Neural Networks and Deep Learning

- **Neural Networks (NN)** is a beautiful biologically-inspired machine learning paradigm which enables a computer to learn from observational data.
- **Deep Learning (DL)** is a powerful set of tricks and techniques for learning in neural networks.
- NN and DL currently provide the best solutions to **many problems in image recognition, speech recognition, and natural language processing.**

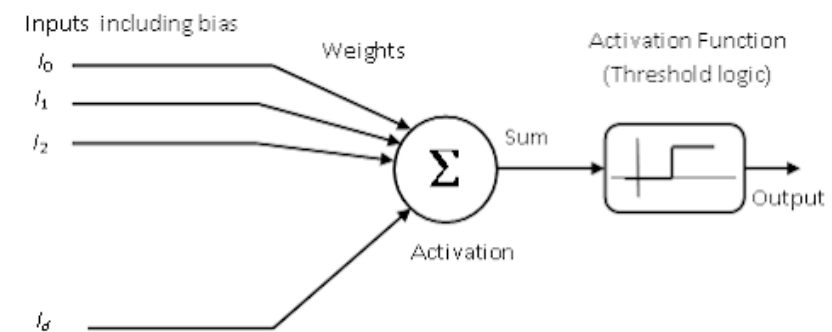
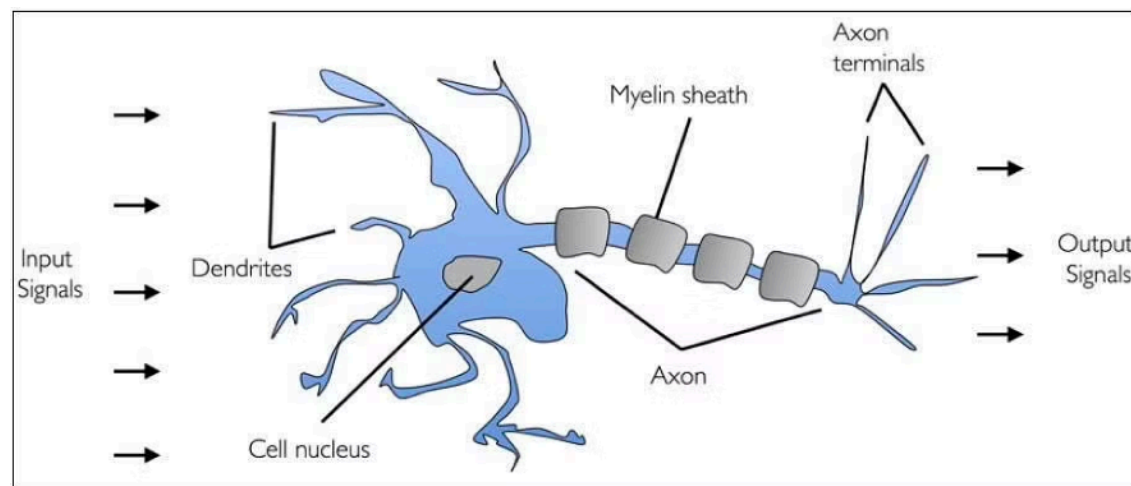
Neural Networks and Deep Learning



In 1943, neurophysiologist **Warren McCulloch** and mathematician **Walter Pitts** wrote a paper on how neurons might work.



They described the brain's activity through simple neural networks built from electrical circuits **capable of performing logical functions**, showing that **neurons can be understood as computing elements**.



$$f_1(\mathbf{x}) = \mathbf{w}\mathbf{x} = \sum_i w_i x_i = x_0 w_0 + \dots + x_m w_m$$

$$f_2(\mathbf{x}) = \begin{cases} 1 & f_1(\mathbf{x}) > 0 \\ 0 & \text{otherwise} \end{cases}$$

Neural Networks and Deep Learning



Donald Hebb was a psychologist who, in 1949, proposed a key theory about how learning happens in the brain. His main contribution was the idea that connections between neurons strengthen when those neurons are active at the same time.

👉 This is often summarized as “cells that fire together, wire together.”



In 1957 **Frank Rosenblatt** attempted to build a kind of mechanical brain called the **Perceptron**, which was billed as “*a machine which senses, recognizes, remembers, and responds like the human mind*”.

Neural Networks and Deep Learning

NEW NAVY DEVICE LEARNS BY DOING

Psychologist Shows Embryo
of Computer Designed to
Read and Grow Wiser

WASHINGTON, July 7 (UPI)—The Navy revealed the embryo of an electronic computer today that it expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.

The embryo—the Weather Bureau's \$2,000,000 "704" computer—learned to differentiate between right and left after fifty attempts in the Navy's demonstration for newsmen.

The service said it would use this principle to build the first of its Perceptron thinking machines that will be able to read and write. It is expected to be finished in about a year at a cost of \$100,000.

Dr. Frank Rosenblatt, designer of the Perceptron, conducted the demonstration. He said the machine would be the first device to think as the human brain. As do human be-

ings, Perceptron will make mistakes at first, but will grow wiser as it gains experience, he said.

Dr. Rosenblatt, a research psychologist at the Cornell Aeronautical Laboratory, Buffalo, said Perceptrons might be fired to the planets as mechanical space explorers.

Without Human Controls

The Navy said the perceptron would be the first non-living mechanism "capable of receiving, recognizing and identifying its surroundings without any human training or control."

The "brain" is designed to remember images and information it has perceived itself. Ordinary computers remember only what is fed into them on punch cards or magnetic tape.

Later Perceptrons will be able to recognize people and call out their names and instantly translate speech in one language to speech or writing in another language, it was predicted.

Mr. Rosenblatt said in principle it would be possible to build brains that could reproduce themselves on an assembly line and which would be conscious of their existence.

In today's demonstration, the "704" was fed two cards, one with squares marked on the left side and the other with squares on the right side.

Learns by Doing

In the first fifty trials, the machine made no distinction between them. It then started registering a "Q" for the left squares and "O" for the right squares.

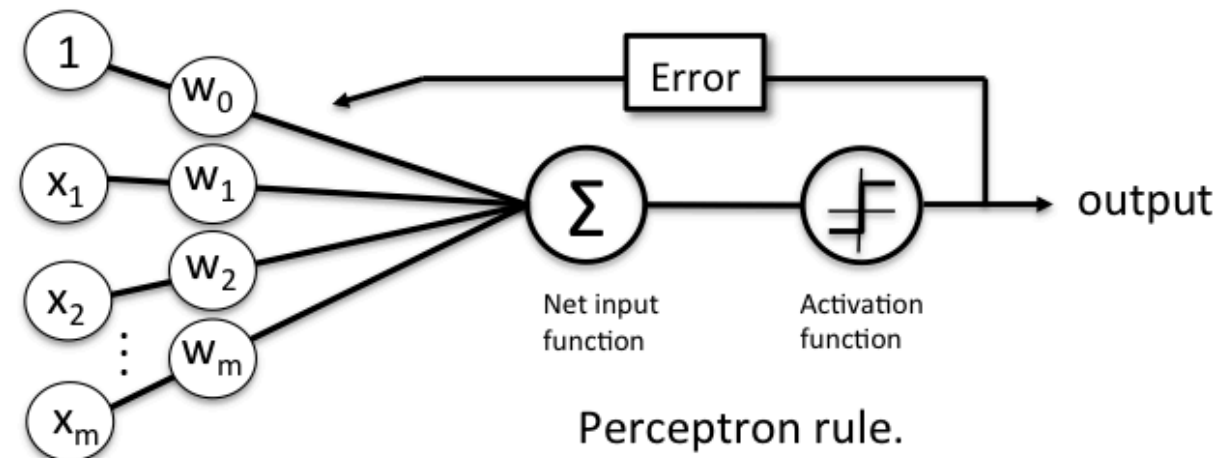
Dr. Rosenblatt said he could explain why the machine learned only in highly technical terms. But he said the computer had undergone a "self-induced change in the wiring diagram."

The first Perceptron will have about 1,000 electronic "association cells" receiving electrical impulses from an eye-like scanning device with 400 photo-cells. The human brain has 10,000,000,000 responsive cells, including 100,000,000 connections with the eyes.

Original article of The New York Times describing the official presentation of the Perceptron by Rosenblatt and his staff

Neural Networks and Deep Learning

The perceptron training rule



Given a dataset of samples that represents the desired behavior of the neuron, we can learn this behavior by following this algorithm:

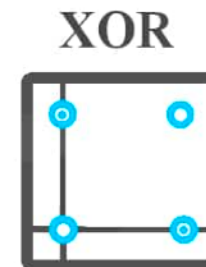
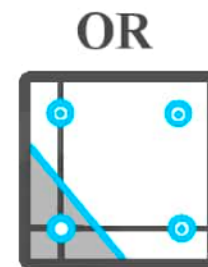
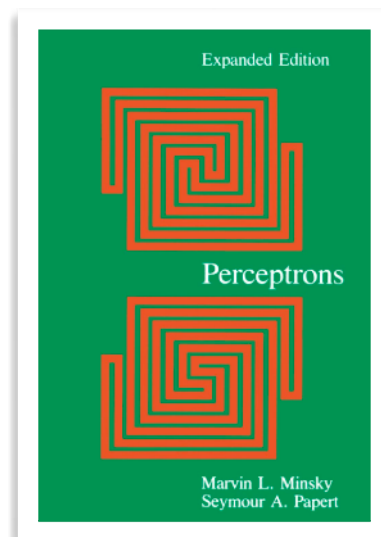
Algorithm: Perceptron Learning Algorithm

```
 $P \leftarrow \text{inputs with label } 1;$   
 $N \leftarrow \text{inputs with label } 0;$   
Initialize  $\mathbf{w}$  randomly;  
while !convergence do  
    Pick random  $\mathbf{x} \in P \cup N$  ;  
    if  $\mathbf{x} \in P$  and  $\mathbf{w} \cdot \mathbf{x} < 0$  then  
        |  $\mathbf{w} = \mathbf{w} + \mathbf{x}$  ;  
    end  
    if  $\mathbf{x} \in N$  and  $\mathbf{w} \cdot \mathbf{x} \geq 0$  then  
        |  $\mathbf{w} = \mathbf{w} - \mathbf{x}$  ;  
    end  
end  
//the algorithm converges when all the  
inputs are classified correctly
```

The Perceptron learning algorithm is guaranteed to converge only if the data is linearly separable. Otherwise, it may cycle forever without finding a solution.

Neural Networks and Deep Learning

A critical book written in 1969 by **Marvin Minsky** and his collaborator **Seymour Papert** (“Perceptrons”) showed that Rosenblatt’s original system was painfully **limited**, literally blind to some simple logical functions like “exclusive-or”. What had become known as the field of “neural networks” almost disappeared.



The Perceptron learning algorithm is guaranteed to converge only if the data is linearly separable. Otherwise, it may cycle forever without finding a solution.

Obviously, there was a **blind spot** in their analysis, but they started the first **neural network winter**.

Neural Networks and Deep Learning



In 1982, interest in the field was renewed. **John Hopfield** of Caltech presented a paper to the National Academy of Sciences. His approach was to create more useful machines by using **bidirectional lines**. Previously, the connections between neurons was only one way.



In 1986, three independent groups of researchers, which included **David E. Rumelhart**, **Geoffrey E. Hinton** and **Ronald J. Williams**, came up with ideas which are now called **back-propagation** networks because it distributes pattern recognition errors throughout the network.



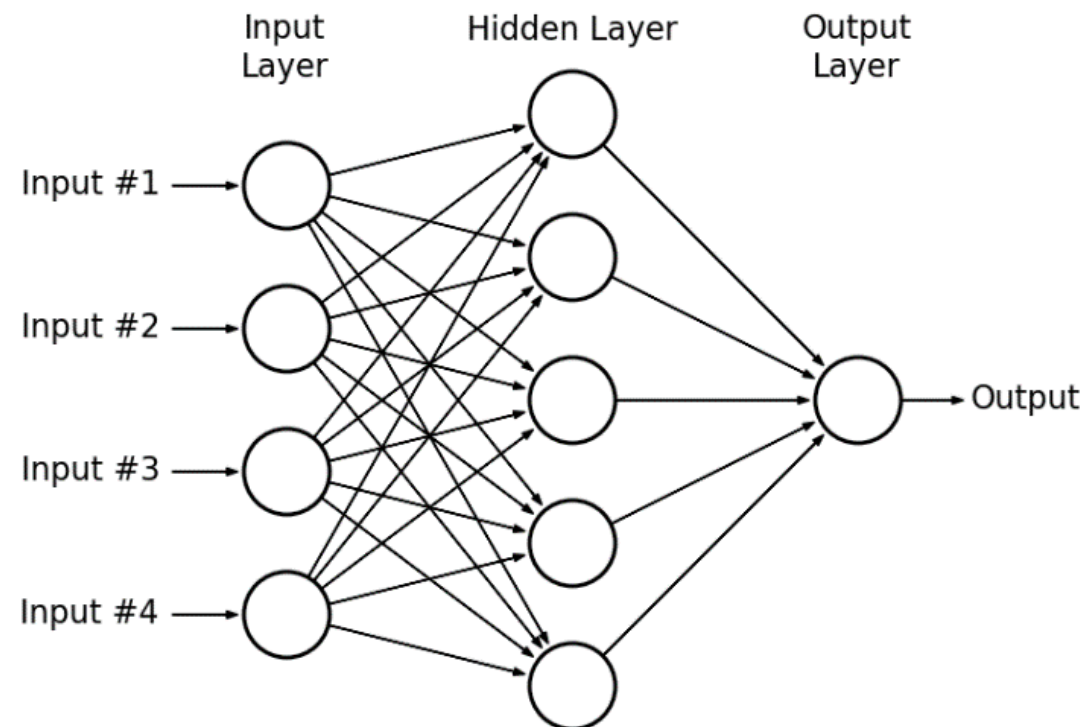
From 1986 to mid 90's new developments arised: **convolutional neural networks** (**Y.LeCun**), **unsupervised learning** (**Y.Bengio**), **RBM** (**G.Hinton**), etc.



But, by this point **new machine learning methods** had begun to also emerge, and people were again beginning to be **skeptical of neural nets** since they seemed so intuition-based and since computers were still barely able to meet their computational needs.

Neural Networks and Deep Learning

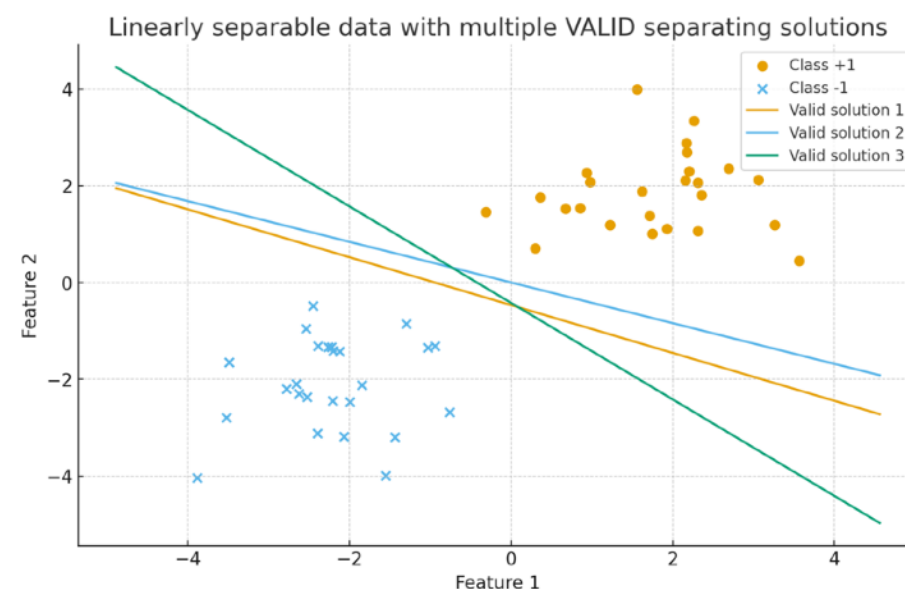
Backpropagation generalized the perceptron training algorithm to several layers (that can solve the XOR problem), tackling the problem raised by Minsky and Papert.



Neural Networks and Deep Learning

Nevertheless, it still had to limitations:

- While the backpropagation algorithm is guaranteed to converge on some solution in the case of a linearly separable training set, it may still pick **any solution** and problems may admit many valid solutions of varying quality.



- For networks with more than two layers, the algorithm's convergence was extremely slow and could even stall or “die,” causing the network to remain undertrained.

Neural Networks and Deep Learning

With the ascent of **Support Vector Machines** and the failure of backpropagation, the early 2000s were a dark time for neural net research (**second neural network winter**).

After that, following Moore's law, computers got dozens of times faster (**GPUs**) since the slow days of the 90s, making learning with large datasets and many layers much more tractable.

Finally, in 2012, the breakthrough came when Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton introduced **AlexNet**, a deep convolutional neural network, which dramatically outperformed previous methods in the ImageNet competition. This success marked the resurgence of neural networks and deep learning.

Neural Networks and Deep Learning



Geological formation, formation
(geology) the geological features of the earth

1808 pictures 86.24% Popularity Percentile Wordnet IDs

Numbers in brackets: (the number of synsets in the subtree).

ImageNet 2011 Fall Release (32326)

- plant, flora, plant life (4486)
- geological formation, formation (1808)
 - aquifer (0)
 - beach (1)
 - cave (3)
 - cliff, drop, drop-off (2)
 - delta (0)
 - diapir (0)
 - folium (0)
 - foreshore (0)
 - ice mass (10)
 - lakefront (0)
 - massif (0)
 - monocline (0)
 - mouth (0)
 - natural depression, depression (0)
 - natural elevation, elevation (41)
 - oceanfront (0)
 - range, mountain range, range of mountains (0)
 - relict (0)
 - ridge, ridgeline (2)
 - ridge (0)
 - shore (7)
 - slope, incline, side (17)
 - spring, fountain, outflow, outpouring (0)
 - talus, scree (0)
 - vein, mineral vein (1)
 - volcanic crater, crater (2)
 - wall (0)

Treemap Visualization **Images of the Synset** **Downloads**

ImageNet 2011 Fall Release > Geological formation, formation

Natural 	Slope 	Shore
Natural 	Ice 	Water
Natural 	Massif 	Vein
Natural 	Mouth 	Delta
Natural 	Lakefront 	Foreshore
Natural 	Wall 	Talus
Natural 	Monocline 	Volcanic
Natural 	Oceanfront 	Beach
Natural 	Aquifer 	Diapir
Natural 	Cave 	Cliff
Natural 	Spring 	Ridge

Neural Networks and Deep Learning



Neural Networks and Deep Learning

The image shows a screenshot of the Dyson website's header and a product page for the Robot Dyson 360 Eye. The header is dark with the Dyson logo on the left. To the right of the logo is a navigation menu with links: 'Tienda', 'Aspiradoras', 'Ventiladores y Calefactores', 'Airblade™', 'Mi cuenta', and 'Soporte'. Further right is a globe icon. Below the header, the product name 'Robot Dyson 360 Eye™' is displayed in a large, light-colored font. To the right of the product name is a yellow button with the text 'Sea el primero en disfrutarlo'. Below the product name, the text 'El nuevo robot aspirador de Dyson' is written in a smaller, light-colored font. In the center of the page is a large, detailed image of the Robot Dyson 360 Eye, a cylindrical robot with a blue top and a transparent body showing internal components. On the left side of the page, there is a circular video thumbnail with a play button icon. Above the thumbnail, the text 'Vea a James Dyson presentando el nuevo Dyson 360 Eye™ en Tokio' is written in a light-colored font.

dyson

Tienda Aspiradoras Ventiladores y Calefactores Airblade™ Mi cuenta Soporte

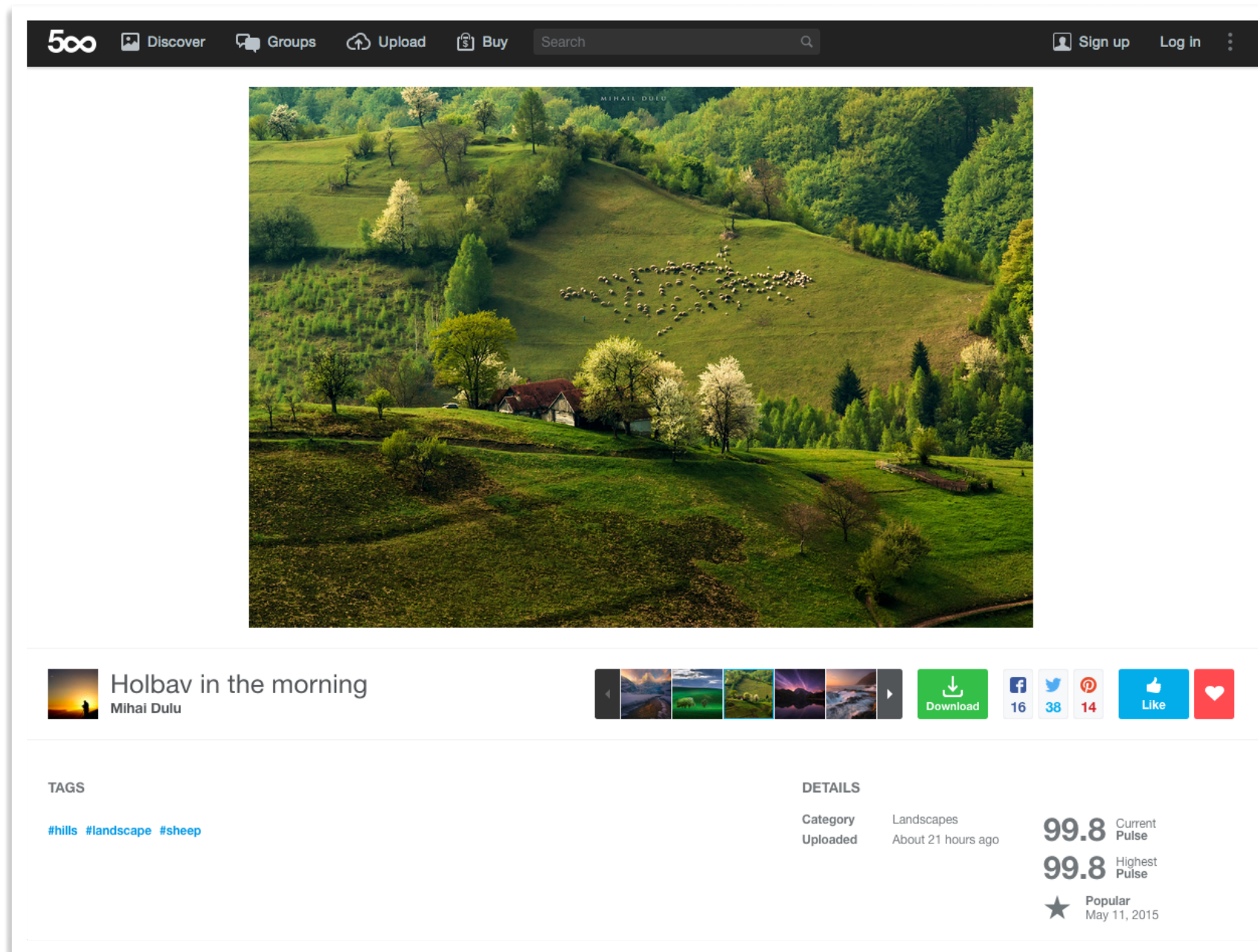
Robot Dyson 360 Eye™

Sea el primero en disfrutarlo

El nuevo robot aspirador de Dyson

Vea a James Dyson presentando el nuevo Dyson 360 Eye™ en Tokio

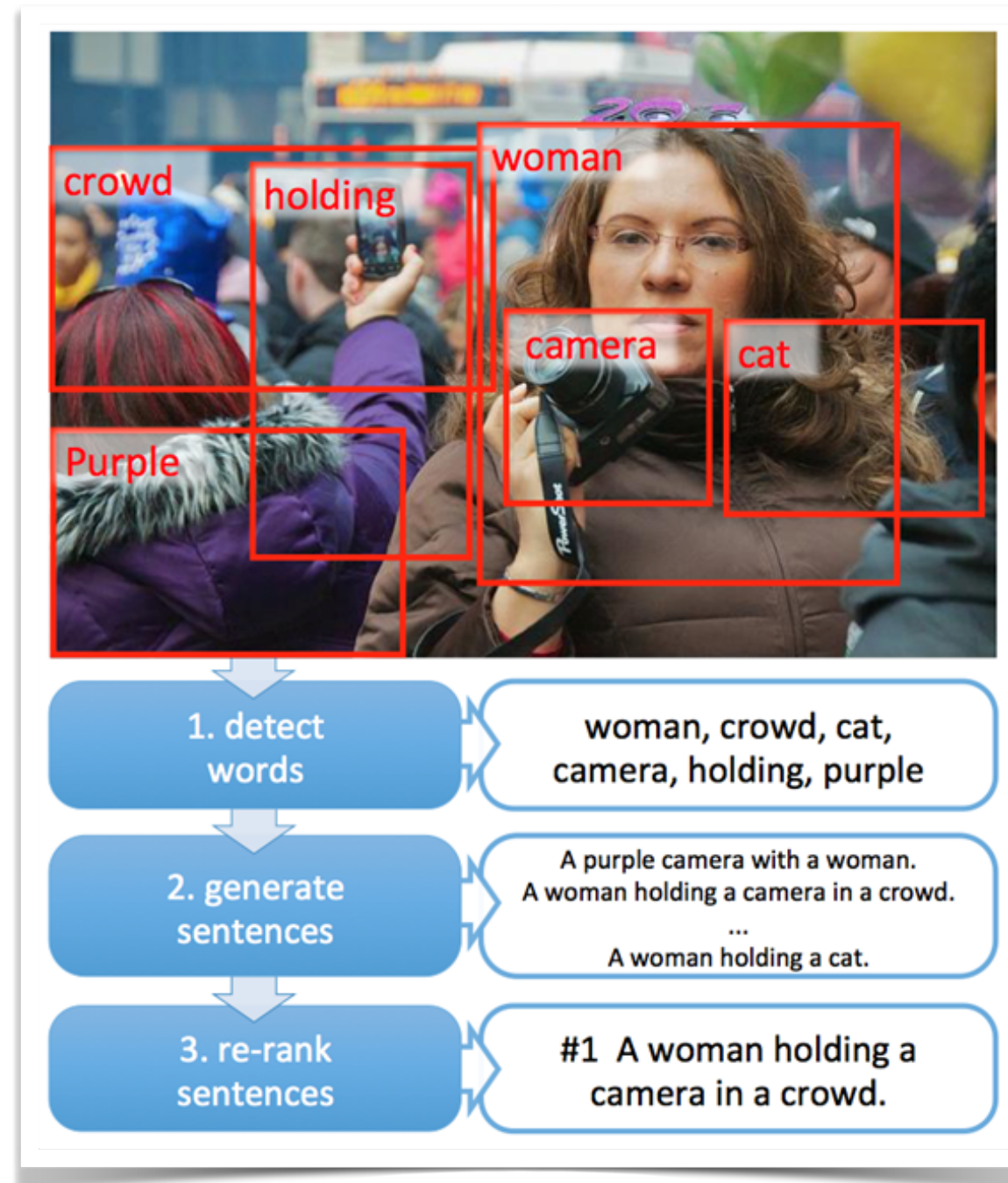
Neural Networks and Deep Learning



What is Pulse?

Pulse is a score out of 100 points that measures how **popular** a photo is. Pulse is calculated by an algorithm, which is unique to 500px and is based on votes (Likes & Favorites) on your photo from the community. The Pulse algorithm was designed to promote daily exposure of new photographs and photographers. It is not necessarily a measure of photograph's quality.

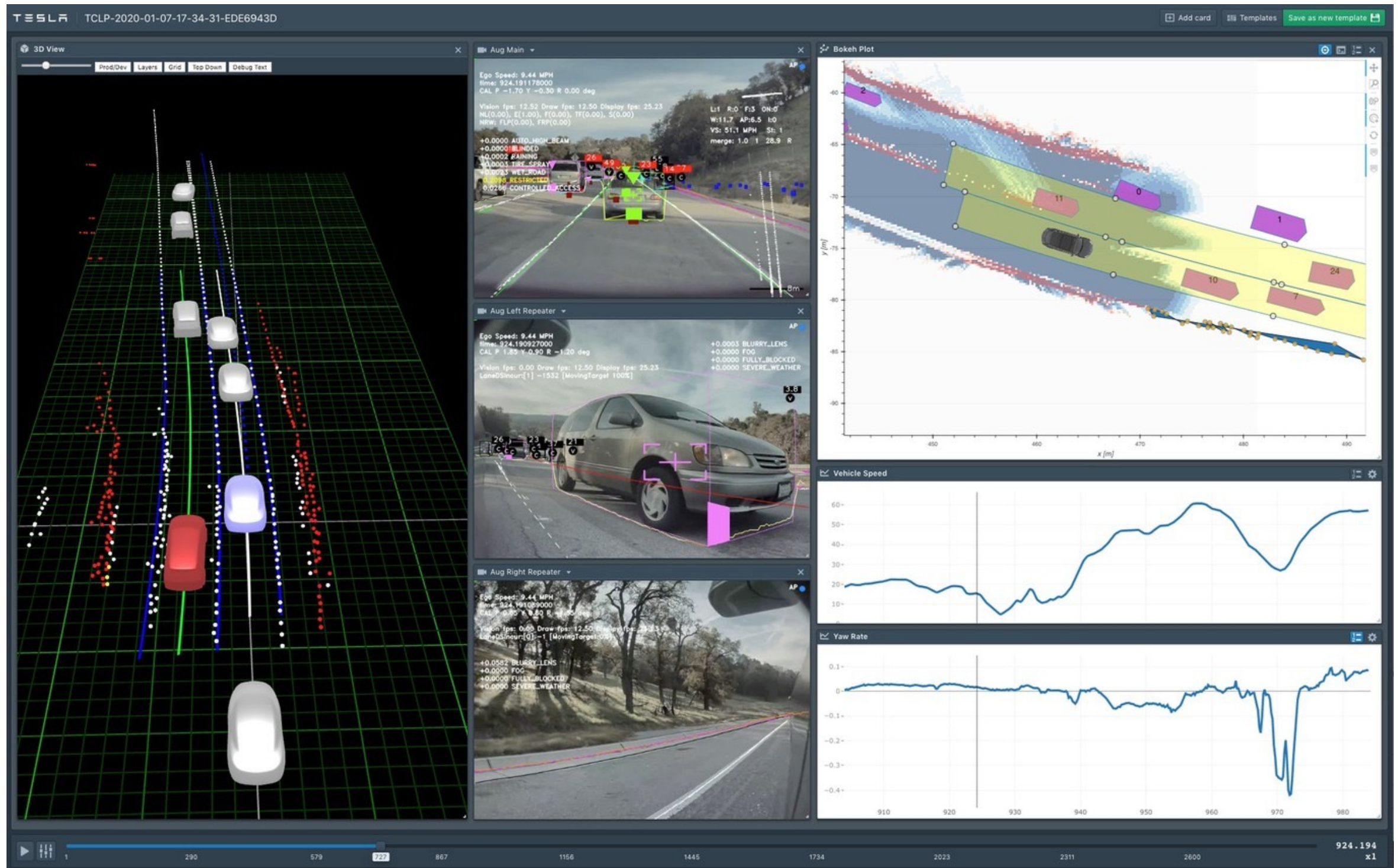
Neural Networks and Deep Learning



Neural Networks and Deep Learning

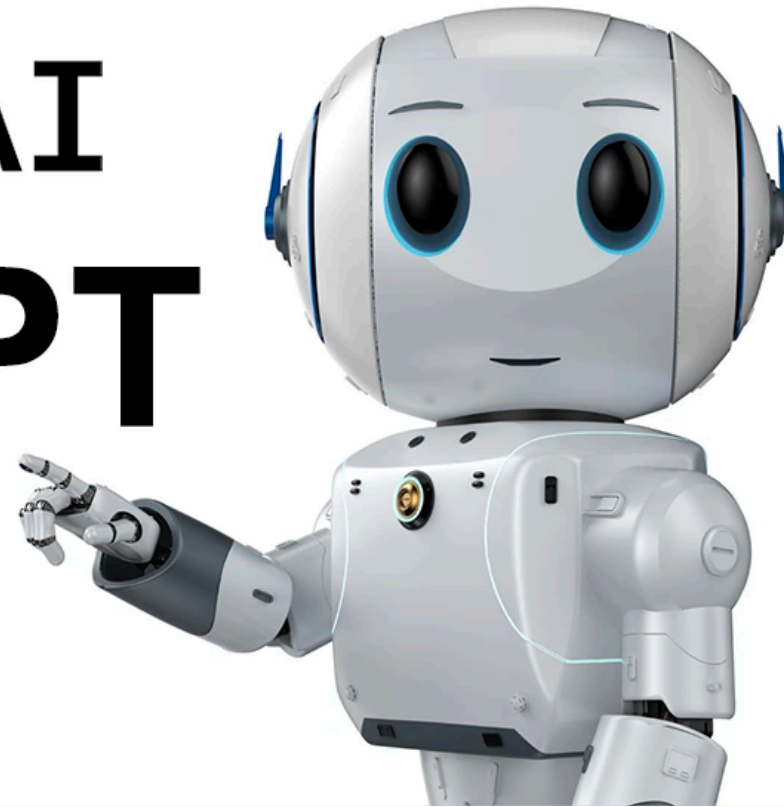


Neural Networks and Deep Learning

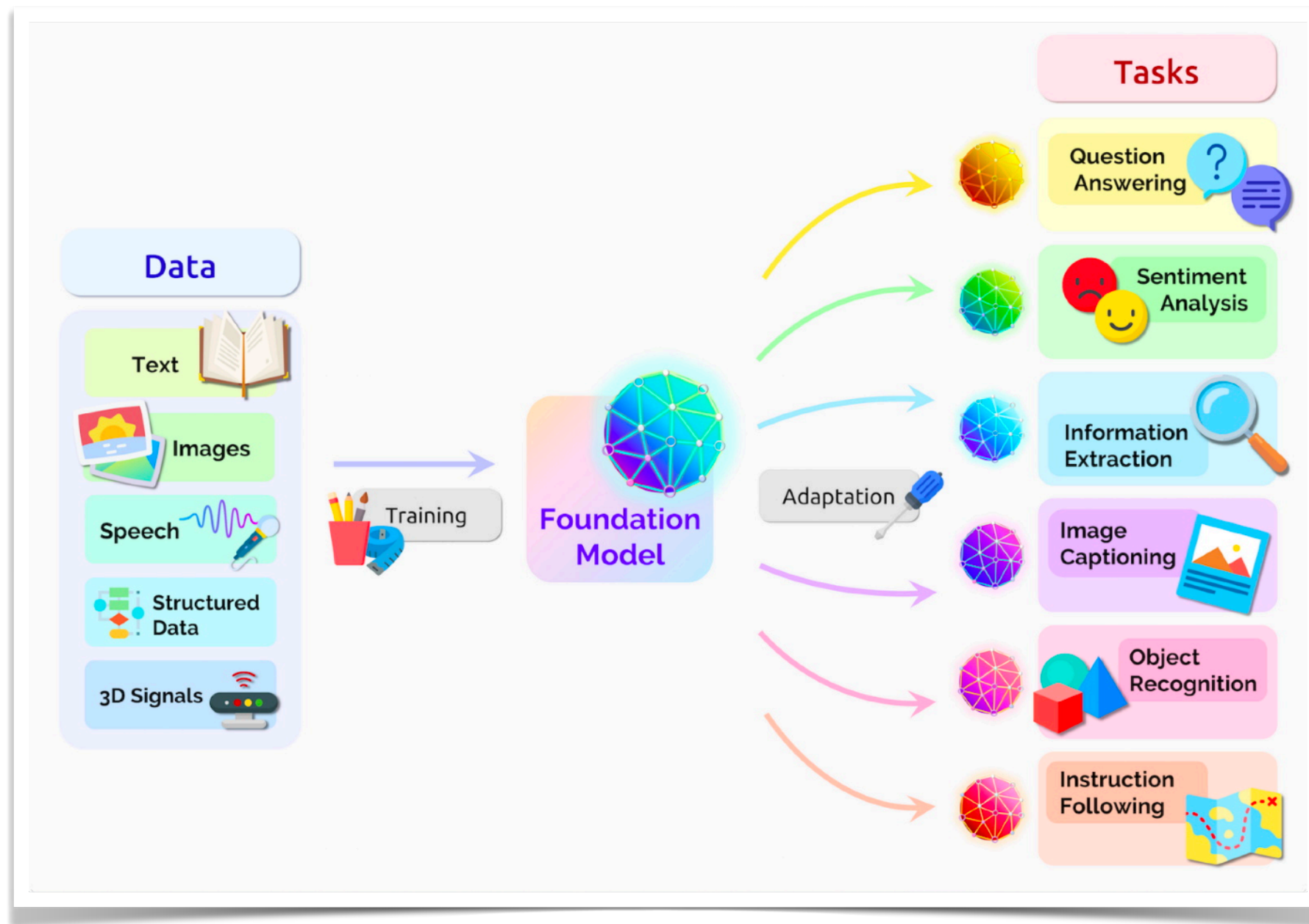


Neural Networks and Deep Learning

 OpenAI
ChatGPT



Neural Networks and Deep Learning

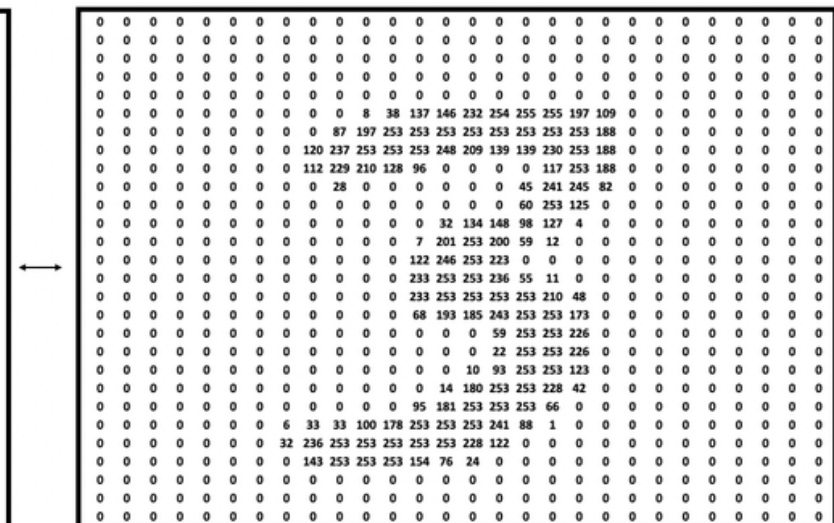
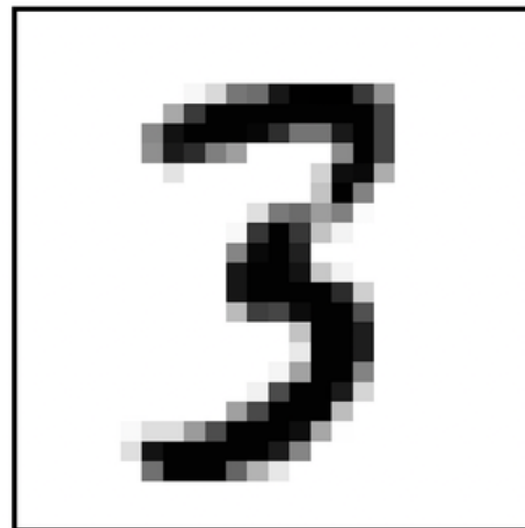
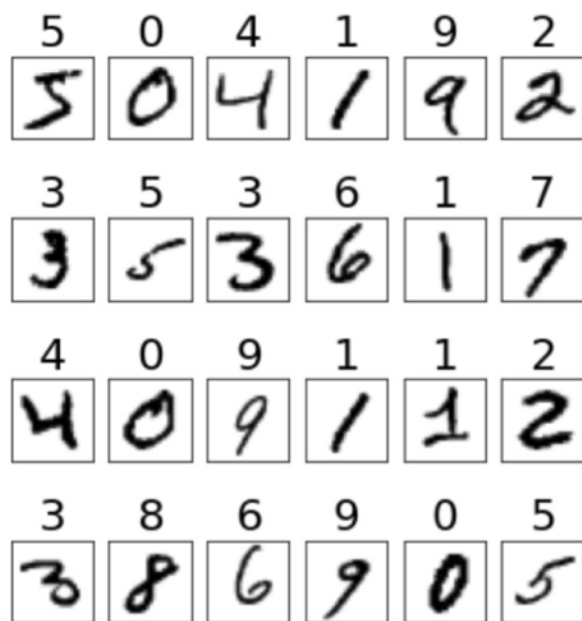


What does it mean to learn from data?

Training data: a set of $(x^{(m)}, y^{(m)})$ pairs.

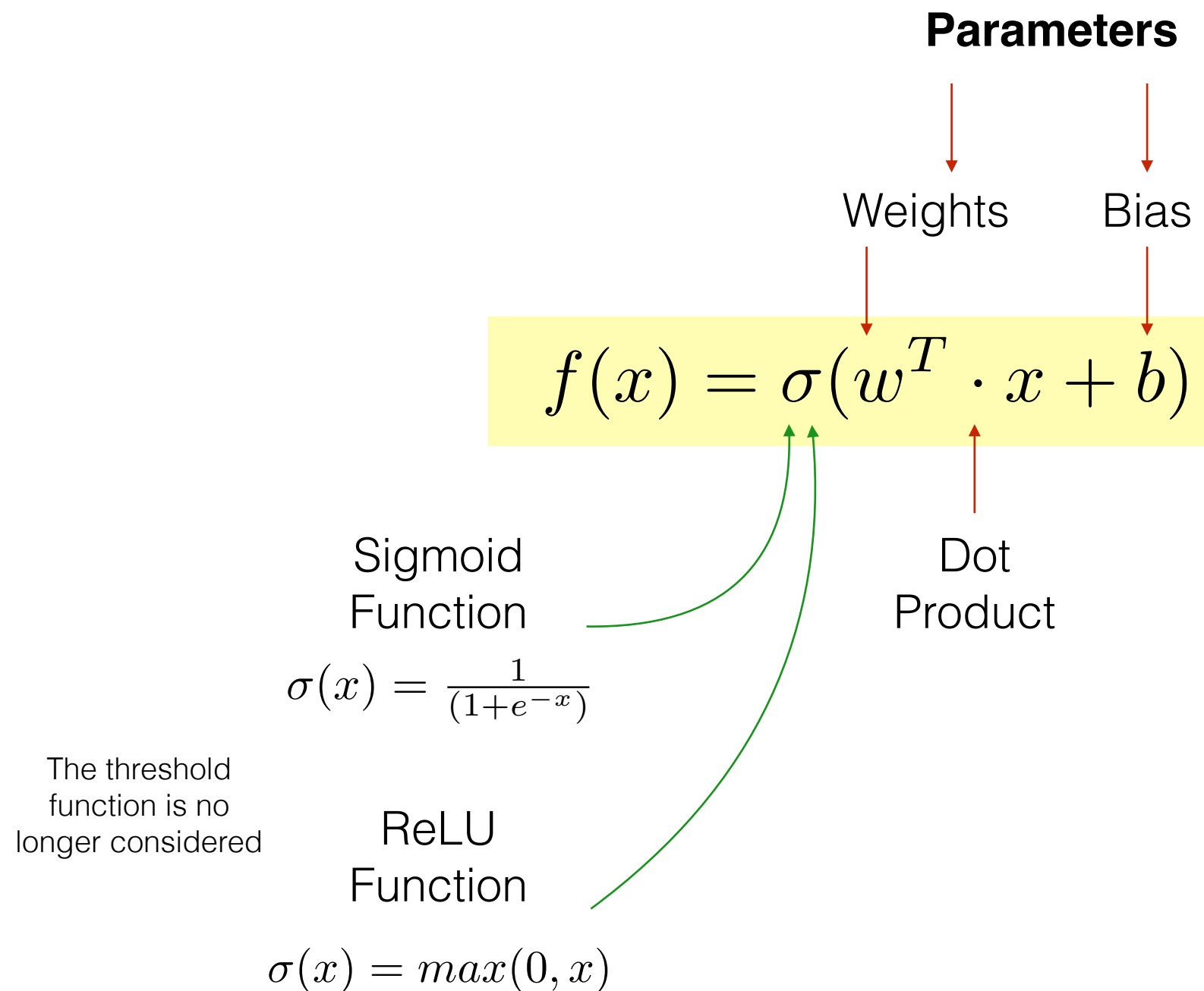
Learn a function $f_w : x \rightarrow y$ to predict on new inputs x .

1. Choose a model function family f_w .
2. Optimize parameters w .

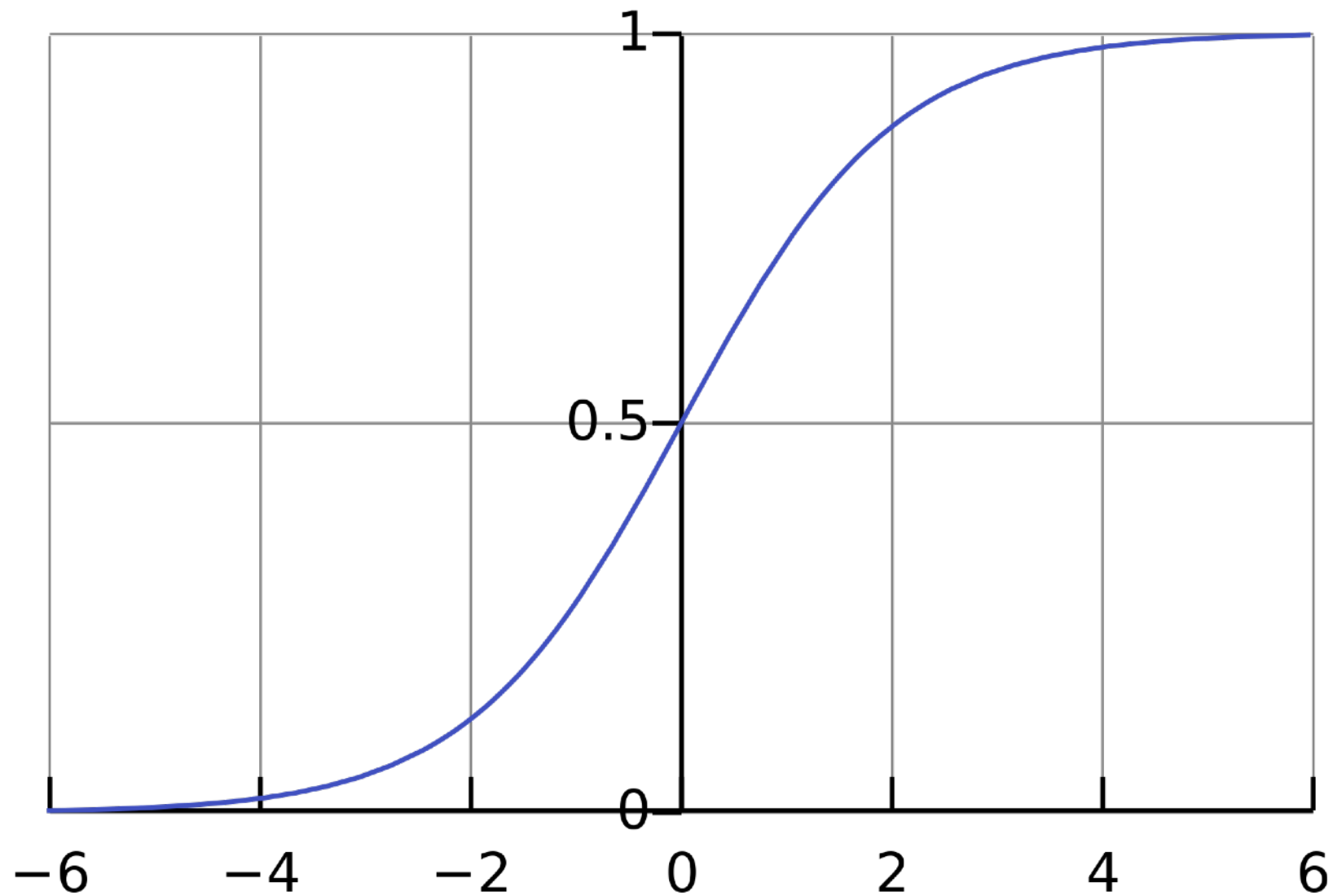


What does it mean to learn from data?

Our first model function: **1-layer “modern” neural net model**

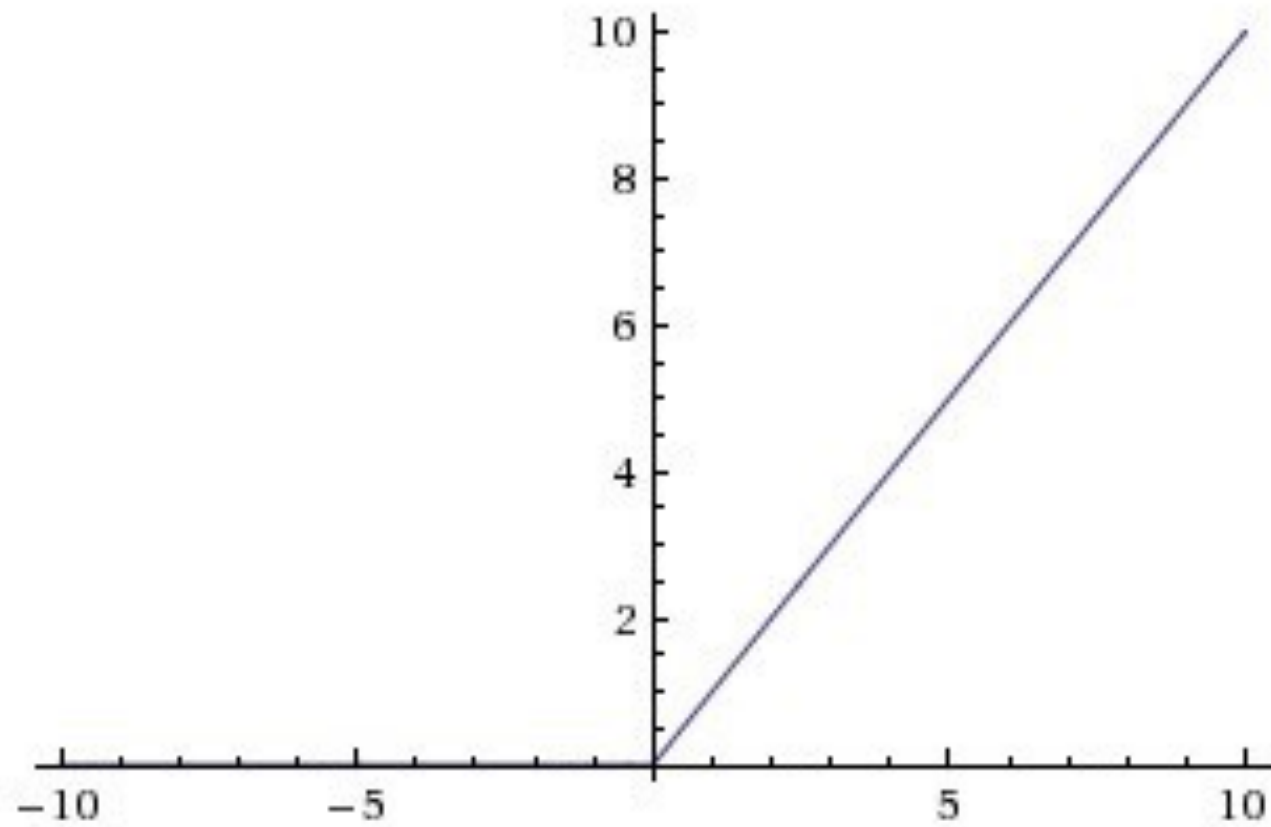


What does it mean to learn from data?



Sigmoid function $\sigma(x) = \frac{1}{(1+e^{-x})}$

What does it mean to learn from data?



ReLU function

$$\sigma(x) = \max(0, x)$$

What does it mean to learn from data?

Table 3: Non-linearities tested.

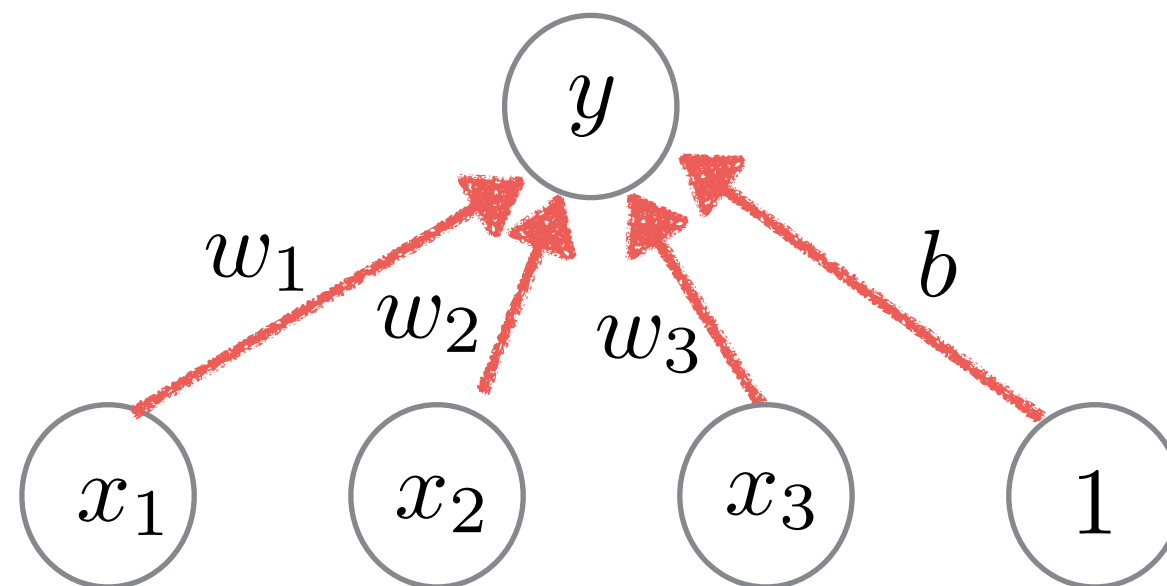
Name	Formula	Year
none	$y = x$	-
sigmoid	$y = \frac{1}{1+e^{-x}}$	1986
tanh	$y = \frac{e^{2x}-1}{e^{2x}+1}$	1986
ReLU	$y = \max(x, 0)$	2010
(centered) SoftPlus	$y = \ln(e^x + 1) - \ln 2$	2011
LReLU	$y = \max(x, \alpha x), \alpha \approx 0.01$	2011
maxout	$y = \max(W_1x + b_1, W_2x + b_2)$	2013
APL	$y = \max(x, 0) + \sum_{s=1}^S a_i^s \max(0, -x + b_i^s)$	2014
VReLU	$y = \max(x, \alpha x), \alpha \in 0.1, 0.5$	2014
RReLU	$y = \max(x, \alpha x), \alpha = \text{random}(0.1, 0.5)$	2015
PReLU	$y = \max(x, \alpha x), \alpha \text{ is learnable}$	2015
ELU	$y = x, \text{ if } x \geq 0, \text{ else } \alpha(e^x - 1)$	2015

What does it mean to learn from data?

1-layer neural net model

$$f(x) = \sigma(w^T \cdot x + b)$$

1-dimensional NN $\rightarrow$ 2 parameters



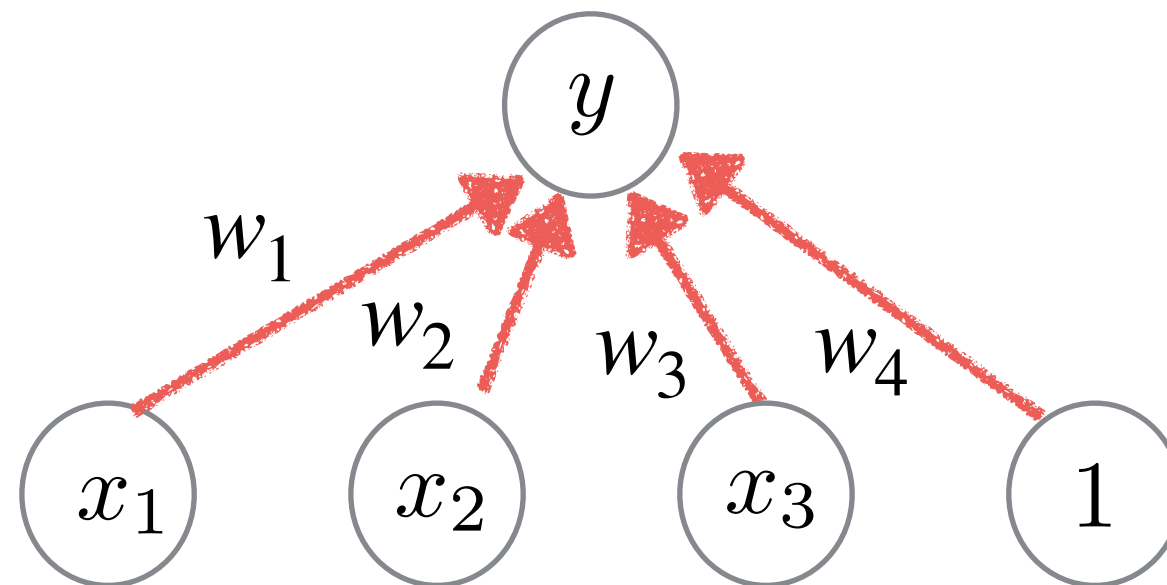
Graphical representation for 3-dimensional data $\rightarrow$ 4 parameters

What does it mean to learn from data?

1-layer neural net model **rewriting**

$$f(x) = \sigma(w^T \cdot x + b) \rightarrow f(\mathbf{x}') = \sigma(\mathbf{w}^T \cdot \mathbf{x}')$$

where $\mathbf{x}'$ is $(x_1, x_2, \dots, x_n, 1)$



In this way we simplify notation and algorithms!

What does it mean to learn from data?

Training a network consist of updating the vector w values until our equation fits the training data as well as possible.

Such updates happen by randomly picking one of the examples and determining if it is a miss-classification. You have a misclassified example when the model determines that an example is part of the class, but it isn't, or when the model determines an example isn't part of the class, but it is.

Training usually handles one misclassified example at a time and operates by changing the w vector using a simple weighted addition. In the case of the perceptron ($y_i \in \{-1, +1\}$):

$$w_i = w_i + \alpha(x_i \cdot y_i)$$

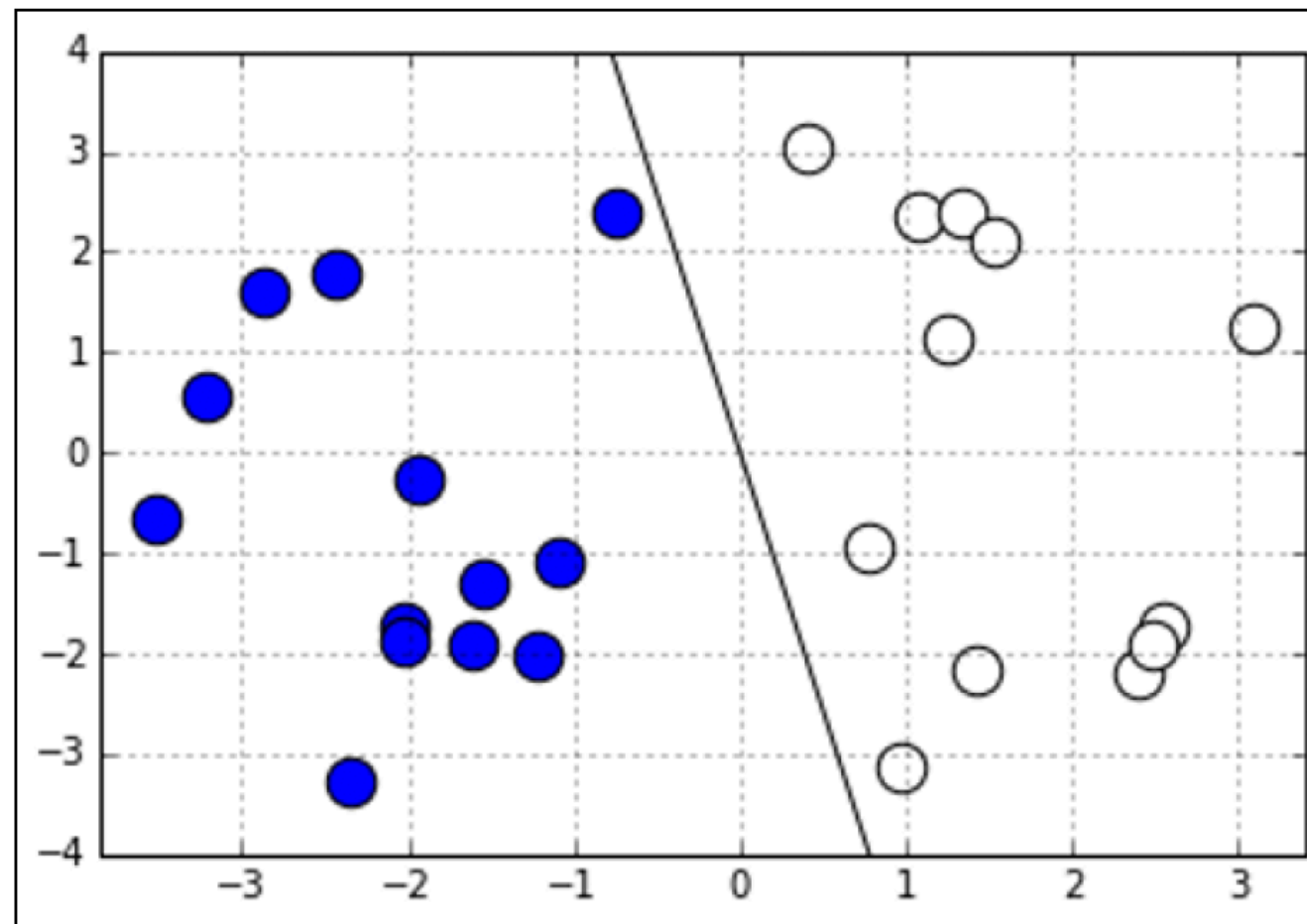
What does it mean to learn from data?

The 2D network is nothing more than a line trying to separate the positive class from the negative one.

Initially, when w is set to zero or to random values, the separating line is just one of the infinite possible lines found on a plane

The updating phase defines it by forcing it to become nearer to the misclassified point. As the algorithm passes through the misclassified examples, it applies a series of corrections. In the end, using multiple iterations to define the errors, the algorithm places the separating line at the border area between the two classes.

What does it mean to learn from data?



This is an example.
Is this an optimal boundary?

What does it mean to learn from data?

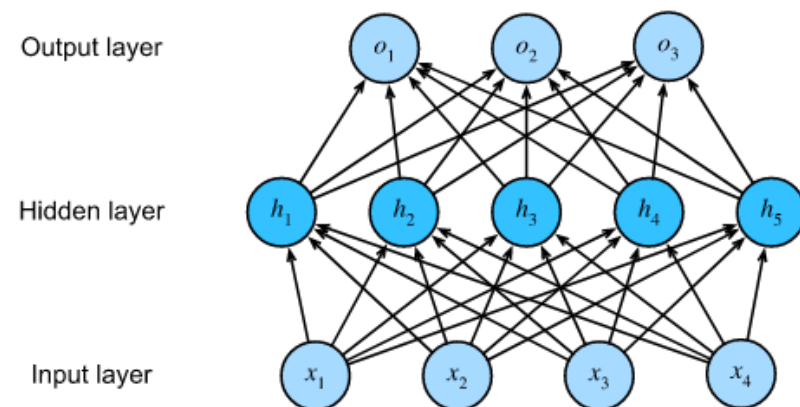
If you can't divide two classes spread on two or more dimensions by any line or plane, they're **nonlinearly separable**. Overcoming data's being nonlinearly separable is one of the challenges of main challenges of ML.

In our case, this can be solved by using **multilayer perceptrons**.

What does it mean to learn from data?

Multilayer (MLP) neural net model

Now we have more parameters!

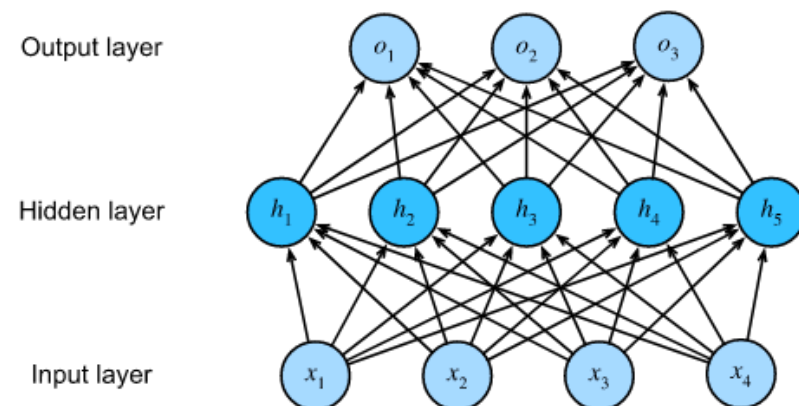


An MLP with a hidden layer of 5 hidden units.

Structure	Types of Decision Regions	Exclusive-or Problem	Classes with Meshed Regions	Region Shapes
One Layer 	Half-Plane			
Two Layers 	Typically Convex			
Three Layers 	Arbitrary			

What does it mean to learn from data?

Multilayer (MLP) neural net model



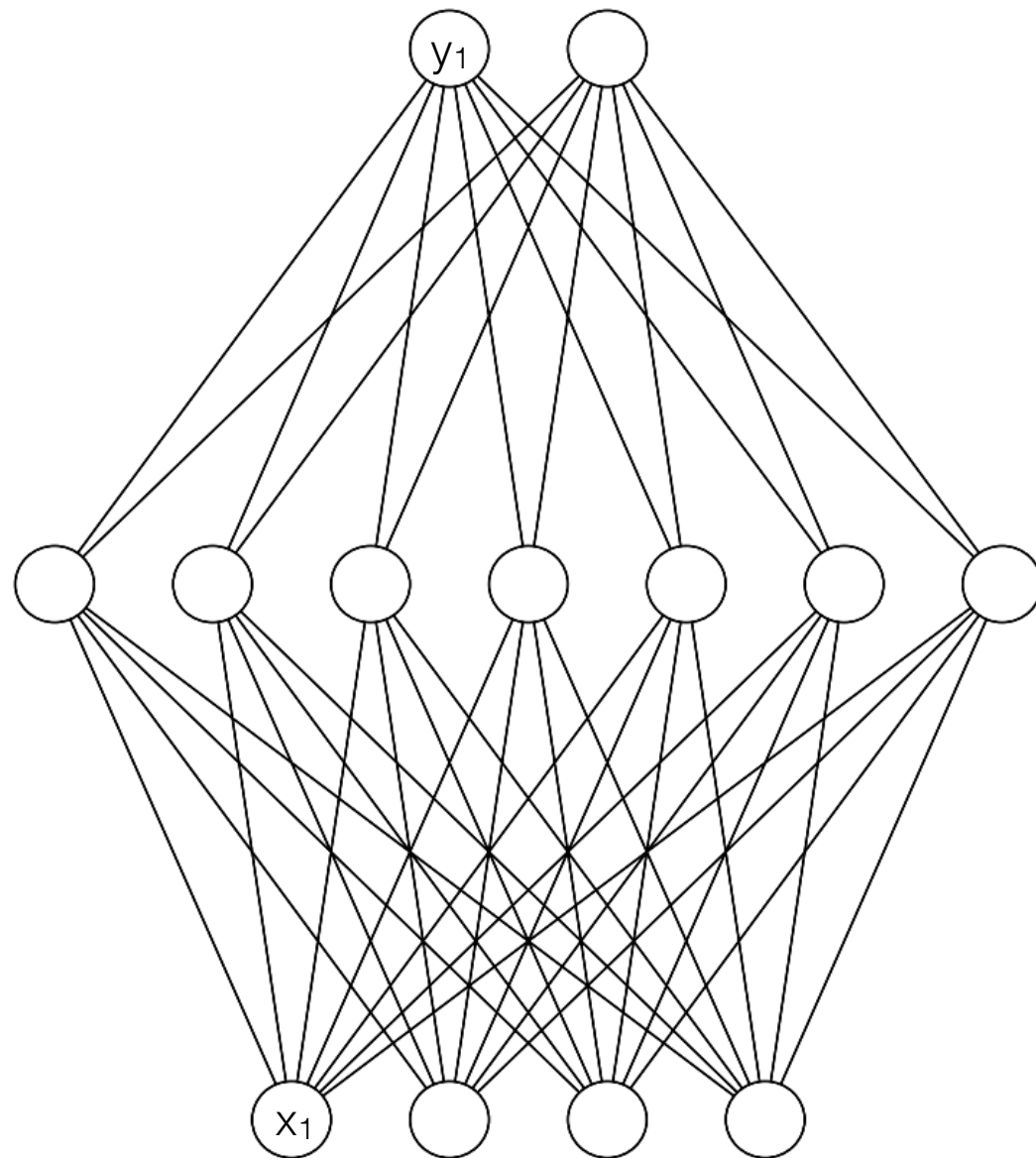
An MLP with a hidden layer of 5 hidden units.

It can be shown (**Universal Approximation Theorem**) that a single-hidden-layer network, given enough nodes (possibly absurdly many), and the right set of weights, we can model any function.

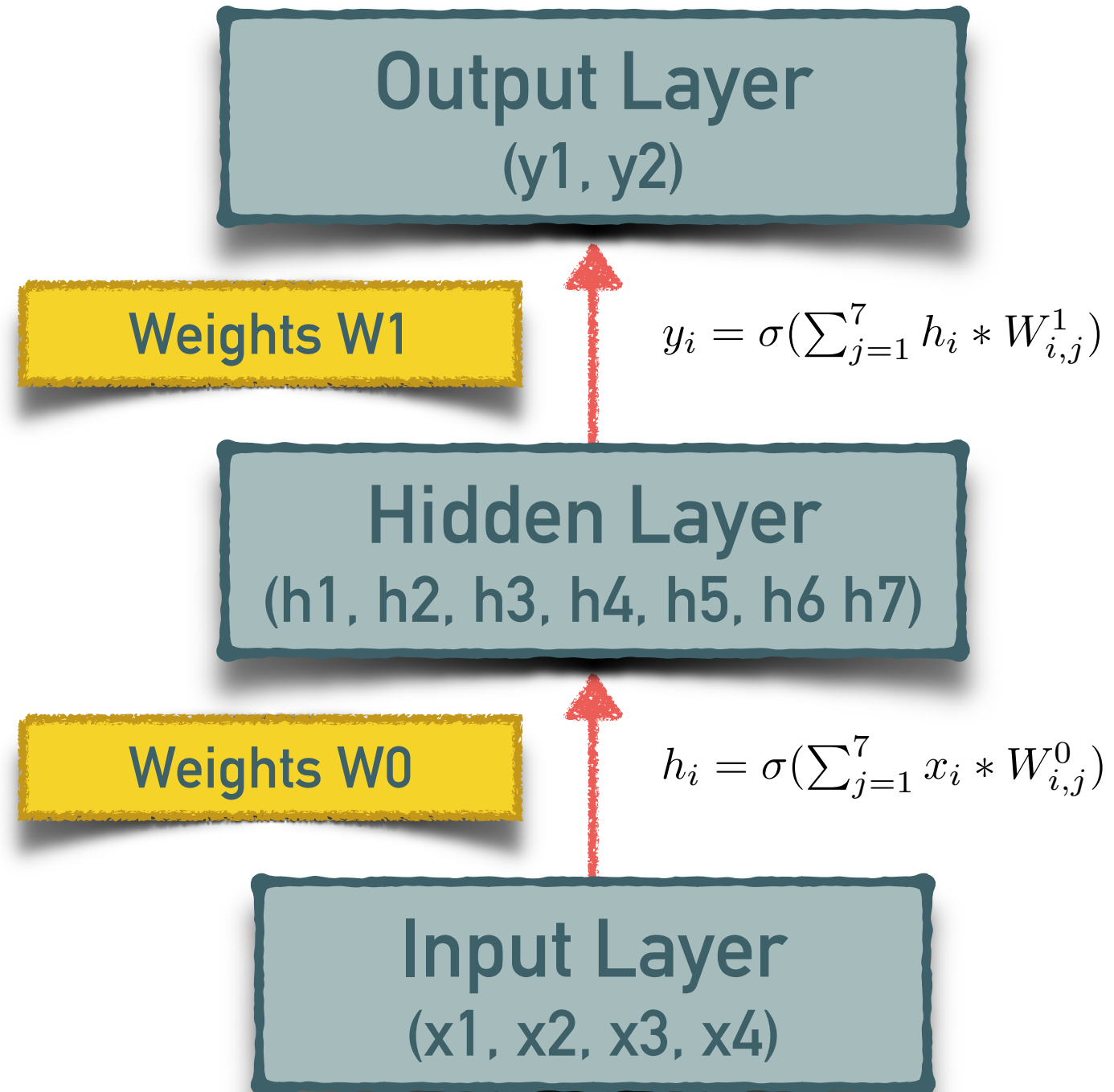
Moreover, just because a single-hidden-layer network can learn any function does not mean that you should try to solve all of your problems with single-hidden-layer networks. In fact, we can approximate many functions much more compactly by using deeper (vs. wider) networks.

What does it mean to learn from data?

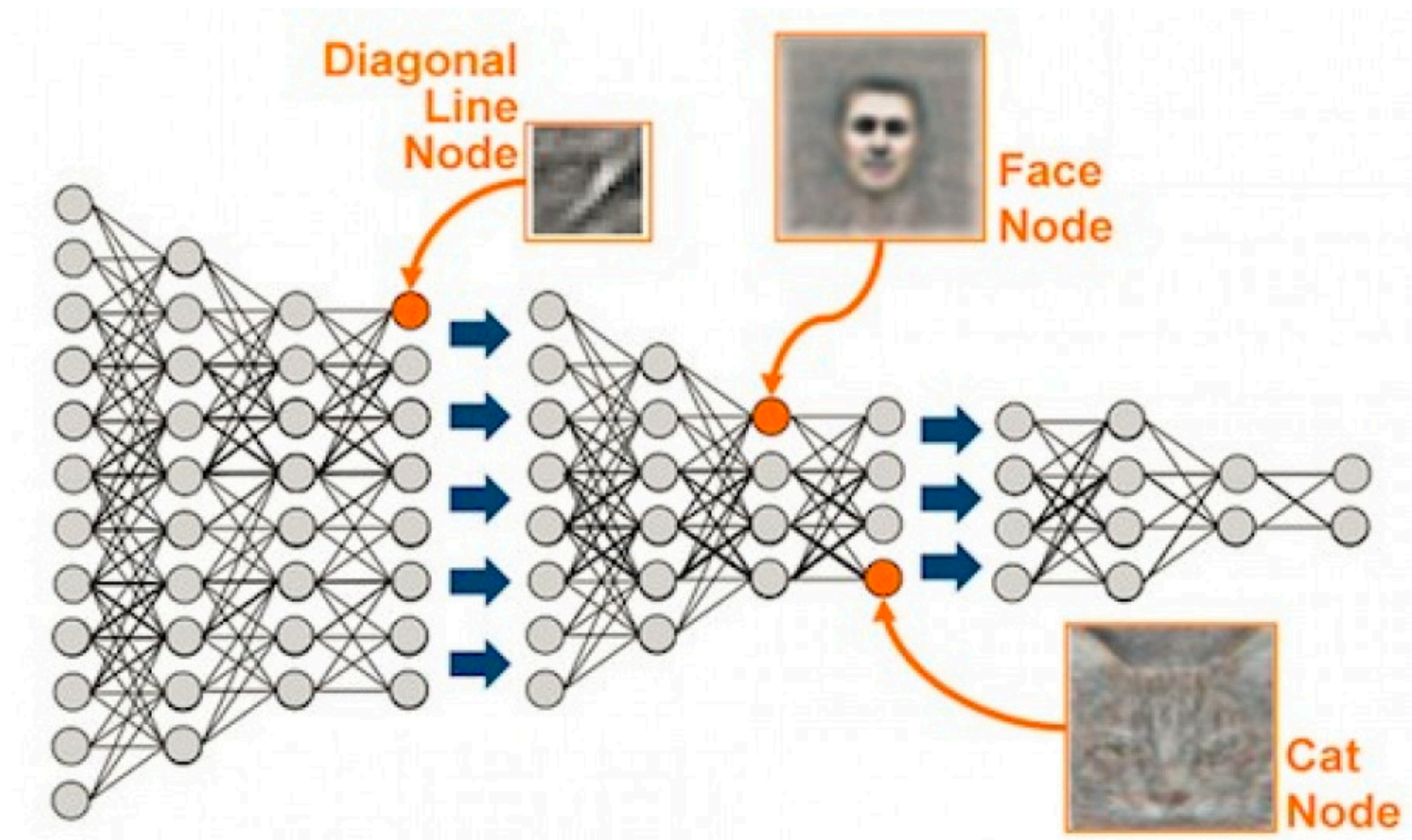
2-layer neural net model



Computing the activation of one layer from the previous one can be written as a matrix-vector multiplication!

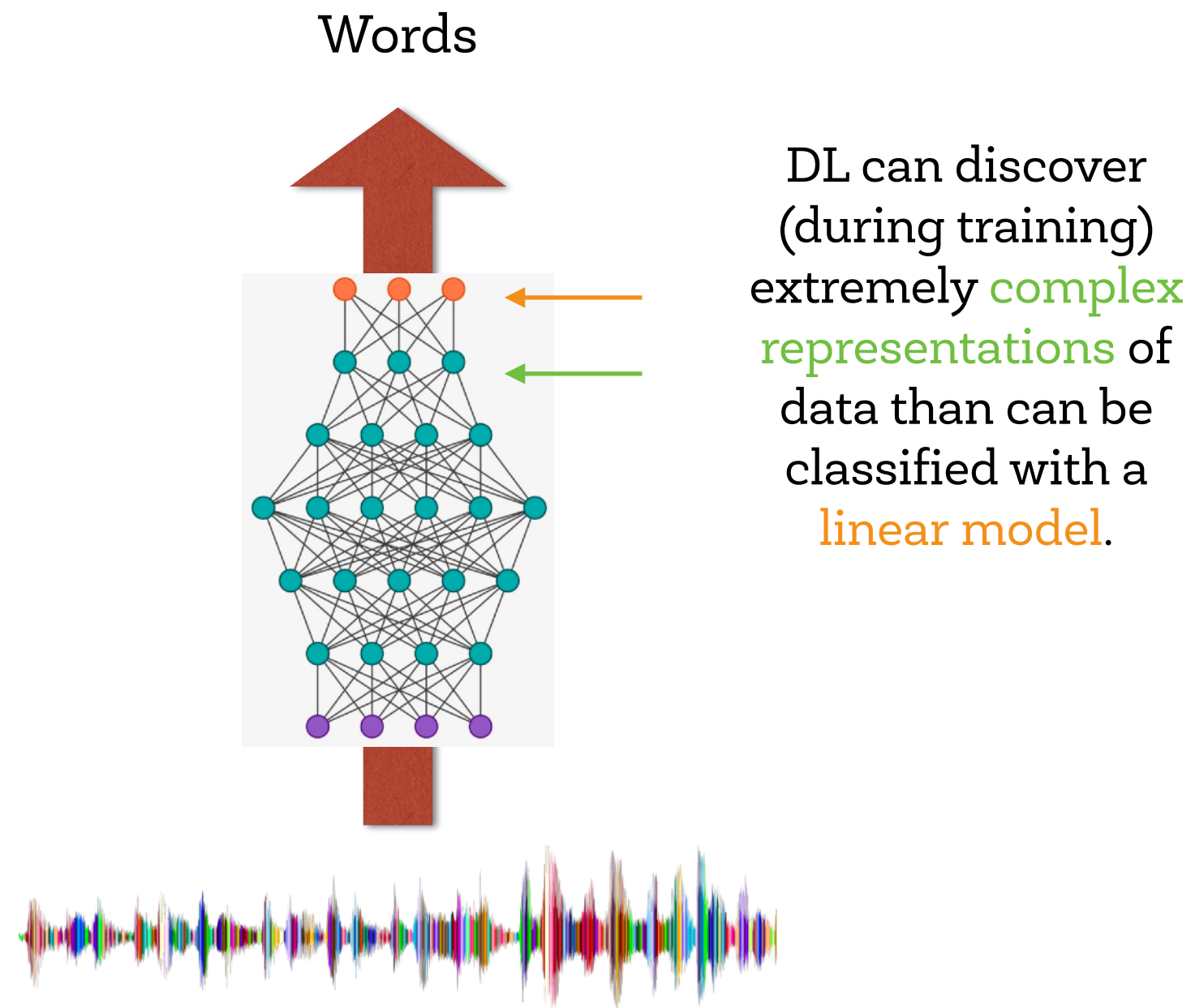


What does it mean to learn from data?



<http://sdat.ir/en/sdat-blog/item/1-deep-learning>

What does it mean to learn from data?



Bibliography

